# BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BANNU

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2021

# PHYSICS

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

## PHYSICS PAPER-10th — ANNUAL EXAM-2021 — SECTION-A

**Time Allowed:** 15 mints | **Maximum Marks:** 12

Note: Q 1: Choose the correct options.

Q1. The electricity deliver of a house in Pakistan has a frequency of 50 Hz. Its time period is.....

(A) 0.2 S
(B) 0.02 S
(C) 0.1 S
(D) 0.015 S

Q2. When the bob of the simple pendulum is at mean position, its P.E is.....

(A) Maximum
(B) half
(C) minimum
(D) zero

Q3. 1 bel is equal to .....

(A) 1 dB
(B) 10 dB
(C) 100 dB
(D) 1000 dB

Q4. When a ray of light falls normally on the surface of mirror then the angle of reflection will be.....

(A) 0°
(B) 30°
(C) 45°
(D) 90°

Q5. The value of angle of deviation does not depend upon.....

(A) angle of prism
(B) angle of incidence.
(C) angle of reflection
(D) refractive index of material of prism

Q6. In parallel combination of capacitors the potential difference across each capacitor is.....

(A) Zero
(B) different
(C) same
(D) negative

Q7. If area of the plates of a capacitor is increased then capacitance of the capacitor will.....

(A) remain the same
(B) increase
(C) decrease
(D) none of these

Q8. Which one is the source of potential difference?

(A) resistor
(B) cell
(C) battery
(D) generator

Q9. The amount of energy obtained by a power of 1 KW in 1 hr. is?

(A) $6.3 \times 10^6$j
(B) $6.3 \times 10^7$j
(C) $3.6 \times 10^6$j
(D) None

Q10. Power is transmitted from a powerhouse on high A.C voltage because .....

(A) of faster transmission
(B) for sure precaution
(C) of less power loss
(D) of power to generate power at low voltage

Q11. .....has high density of data compared to others?

(A) Compact disk
(B) floppy disk
(C) flash drive
(D) hard disk

Q12. Which eyes have higher penetrating power....

(A) x-rays
(B) ɣ-rays
(C) β-rays
(D) α-rays

## SECTION-B

**Time Allowed:** 2.45 hrs. | **Maximum Marks:** 32

Q2. (Marks 32) Q 2: Answer any Eight (8) parts. Each part carries 4 marks.

i. Define the terms: Time period, Frequency, Wavelength, Amplitude.

ii. What is the difference between mechanical and electromagnetic waves? Explain with examples?

iii. How the bats can safely fly at night without colliding with objects?

iv. Why is concave mirror used as shaving mirror? Explain?

v. Write down the hazards of electrostatics?

vi. Give a brief description of electromotive force?

vii. Three resistors each of 5Ω are connected in parallel? Find the equivalent resistance?

viii. Under what circumstances the force on current carrying wire placed in a magnetic field will be maximum?

ix. What is meant by "Electron Gun"? Explain?

x. Define internet? How can it be used for information?

xi. What is half-life of radioactive elements?

## SECTION-C

**Maximum Marks:** 21

> Note: Q3: Attempt any three (3) questions. All carries equal marks.

Q2. Q. 2: i. Define sound? Explain the three things necessary for the production and heating of sound? ii. The time period of an electromagnetic wave is $10^{-15}$. Find its frequency?

Q3. Q. 3: i. What is electrostatic potential? Write an application of electrostatics? ii. How much work is done in moving a charge of 3C from a point at 118V to a point at 138 V?

Q4. Q. 4 i. Describe the working principle, structure and function of a transformer? ii. A transformer has 100 turns in its primary and 500 turns in the secondary coil. If 6V A.C is applied to its primary coil, find the voltage across its secondary co[illegible in source scan]

Q5. Q. 5:i. Define thermionic emission. Explain the deflection of electron beam by electric and magnetic fields? ii. Draw symbol and truth table for AND gate.